Mahatma Gandhi University Kottayam

MGU-UGP (Honours) MGU-BBA (Honours) REGULAR EXAMINATION OCT 2026

Fifth semester

MG5DSECSC304 Design Principle of User Computer Interaction

Duration 2 hours **maximum Marks 70**

Part A – Very Short Answer Questions

Answer all questions. Each question carries 2 marks.

1. List two key advantages of capacitive touchscreens over resistive touchscreens.

1. Multi-touch support: Capacitive screens can recognize multiple touches simultaneously, such as pinch-to-zoom gestures.
2. Better sensitivity and clarity: They respond to light finger touches and generally provide better screen clarity because they do not require pressure-sensitive layers.

2. Summarize the primary limitation of predictive cognitive models.

The primary limitation of predictive cognitive models is that they assume users behave in a predictable and consistent manner. They may not accurately represent differences in experience, emotions, distractions, learning, and real-world situations.

3. Define cart abandonment in e-commerce interaction design.

Cart abandonment occurs when a customer adds products to an online shopping cart but leaves the website without completing the purchase.

Example: A customer adds a mobile phone to the cart but leaves after seeing unexpected delivery charges.

4. Define Ubiquitous Computing (Ubicomp) as envisioned by Mark Weiser.

Ubiquitous computing, proposed by Mark Weiser, is the concept of embedding computers into everyday objects and environments so that computing becomes available everywhere and operates unobtrusively in the background.

Example: Smart lights that automatically adjust according to room conditions.

5. List two interface design adaptations specifically needed for elderly users.

1. Use large text, buttons, and clear icons to improve readability and ease of interaction.
2. Provide simple navigation, high colour contrast, clear feedback, and sufficient time to complete tasks.

Part B – Short Answer Questions

Answer any 6 questions. Each question carries 5 marks.

6. Discuss the psychological impact of system response time and delay on user performance.

System response time is the time taken by a computer system to respond to a user's action. Response time and delay affect user satisfaction, concentration, efficiency, and confidence.

1. User frustration: Long delays make users impatient and dissatisfied.
2. Loss of concentration: Users may forget their previous actions or lose focus while waiting.
3. Reduced productivity: Slow systems increase task completion time and reduce efficiency.
4. Repeated actions: Users may click buttons repeatedly, assuming the system has not received their input.
5. Reduced trust: Frequent delays make users believe the system is unreliable.

Design solutions:

- Display a progress indicator during lengthy operations.
- Provide immediate feedback after every action.
- Optimize the system to reduce unnecessary delays.
- Show timeout messages and retry options when appropriate.

Conclusion: Fast and predictable responses improve user performance, confidence, and satisfaction.

7. Compare remote asynchronous usability testing with moderated in-person usability testing.

Usability testing evaluates how easily users can interact with a system.

Remote asynchronous testing	Moderated in-person testing
Participants complete tasks remotely at their convenience.	Participants complete tasks under the guidance of a moderator.
No moderator is present during the session.	A moderator observes and guides the session.

Less expensive and easier to conduct with many users.	More expensive and requires scheduling.
Suitable for geographically distributed participants.	Usually conducted in a laboratory or a common location.
Limited opportunity to ask follow-up questions immediately.	The moderator can ask questions and clarify user behaviour.
Useful for measuring task completion rates and time.	Useful for understanding user emotions, confusion, and thought processes.

Conclusion: Remote asynchronous testing is suitable for large-scale testing at low cost, whereas moderated in-person testing provides deeper insights into user behaviour.

8. Show how spatial audio can be employed in virtual meeting rooms to improve conversational clarity.

Spatial audio creates the perception that different speakers are located at different positions in a virtual environment.

Applications:

1. Speaker positioning: Each participant's voice appears to come from a different virtual location.
2. Improved identification: Users can identify who is speaking based on the apparent direction of the voice.
3. Reduced confusion: Separating speakers spatially makes overlapping conversations easier to follow.
4. Natural conversation: Virtual meetings become more similar to face-to-face meetings.
5. Adjustable audio: Users can adjust the volume or position of individual speakers when supported.

Example: In a virtual seminar, the presenter may appear in front, while participants are positioned around a virtual table.

Conclusion: Spatial audio improves speaker identification, reduces listening effort, and supports clearer communication in virtual meeting rooms.

9. Implement a dialog flow diagram that handles network timeout and retry semantics in a mobile app.

A mobile application should handle network failures gracefully and allow users to retry unsuccessful operations.

Dialog flow:

Start request

Send request to server
Response received before timeout?
Yes
Display result
No
Show timeout message
Retry or cancel?

Retry → Send request again

Cancel → End operation

Implementation points:

1. Set a reasonable timeout for each network request.
2. Display a message such as "Network timeout. Please try again."
3. Provide Retry and Cancel buttons.
4. Limit automatic retries and use increasing delays between attempts.
5. Prevent duplicate transactions by using idempotency keys or checking the operation status before retrying payments.

Conclusion: Proper timeout and retry handling improves reliability and prevents user confusion and duplicate operations.

10. Show how cognitive workload assessment using the NASA-TLX tool is conducted on an interactive system.

NASA Task Load Index (NASA-TLX) is a subjective tool used to measure a user's perceived workload while performing a task.

It assesses six dimensions:

1. Mental demand: Amount of thinking and concentration required.
2. Physical demand: Physical effort needed to complete the task.
3. Temporal demand: Pressure caused by time limitations.
4. Performance: How successful the user feels they were.
5. Effort: Amount of mental and physical effort required.
6. Frustration: Feelings of stress, irritation, or insecurity.

Procedure:

- Ask participants to perform a specified task on the interactive system.
- After completing the task, ask them to rate each dimension, commonly on a scale from 0 to 100.
- In the traditional NASA-TLX procedure, participants also make pairwise comparisons to assign relative weights to the dimensions.

- Calculate the overall weighted workload score.
- Compare scores across users, tasks, or interface designs.

Example: If users report high mental demand and frustration while completing an online booking form, the designer can simplify the form and reduce unnecessary steps.

Conclusion: NASA-TLX helps designers identify demanding tasks and improve usability.

11. Classify the various human sensory modalities according to their bandwidth and feedback speed in HCI.

Human sensory modalities are the channels through which people receive information from a computer system.

Sensory modality	Bandwidth and feedback characteristics	HCI application
Visual	High information capacity; suitable for complex information.	Text, images, charts, icons and animations.
Auditory	Moderate-to-high information capacity for sequential information; rapid alerts.	Voice instructions, notifications and spatial audio.
Tactile (touch)	Useful for immediate physical feedback; relatively limited information capacity.	Touchscreens, vibration and haptic feedback.
Proprioceptive	Provides information about body position and movement.	Gesture recognition and motion-based interfaces.
Vestibular	Provides information about balance and movement.	VR motion interaction and movement simulation.

Conclusion: Visual and auditory channels are useful for communicating information, while tactile and movement-related senses support direct interaction and feedback. Effective HCI selects the appropriate modality for each task and avoids sensory overload.

12. Implement a floor control protocol for a multi-speaker virtual seminar room.

A floor control protocol manages who is allowed to speak at a particular time in a virtual seminar.

Protocol:

1. Request to speak: A participant clicks the Raise Hand button.
2. Queue management: The system places the participant in a speaking queue.
3. Grant permission: The host or system grants the speaking floor to the next participant.
4. Enable microphone: The participant's microphone is enabled, subject to their device and permissions.
5. Release the floor: After speaking, the participant clicks Done or the host revokes speaking permission.

6. Next speaker: The system grants the floor to the next person in the queue.

Additional features:

- Display the current speaker's name.
- Notify participants when their turn arrives.
- Allow the host to interrupt or remove disruptive speakers.
- Give priority to the moderator for emergencies.
- Prevent two participants from receiving exclusive floor control simultaneously.

Conclusion: A well-designed floor control protocol reduces interruptions, promotes fairness, and makes online seminars easier to manage.

13. Implement Nielsen's "Error prevention" heuristic in the design of a date-picker and flight booking form.

Nielsen's error-prevention heuristic states that interfaces should prevent problems from occurring rather than merely displaying error messages after mistakes happen.

Applications in a date-picker:

1. Disable dates that are in the past when selecting a future flight.
2. Prevent the return date from being earlier than the departure date.
3. Highlight valid dates and clearly distinguish unavailable dates.
4. Use calendar controls instead of requiring users to type dates manually.
5. Validate the selected dates before allowing the user to proceed.

Applications in a flight booking form:

1. Require essential fields such as origin, destination, and departure date.
2. Prevent users from selecting the same airport as both origin and destination, where inappropriate.
3. Restrict passenger numbers to supported limits.
4. Show a booking summary before payment.
5. Ask for confirmation before completing a payment or cancelling a booking.

Example: If a user selects 20 December as the departure date, the system should prevent them from selecting 15 December as the return date.

Conclusion: Error prevention reduces booking mistakes, improves usability, and saves users time and money.

Part C – Essay Questions

Answer any 2 questions. Each question carries 15 marks.

14. Examine the information architecture, navigation paradigms, and interaction design patterns of large-scale social media platforms, analyzing their impact on user engagement and cognitive overload.

Introduction

Social media platforms such as Facebook, Instagram, and YouTube allow users to communicate, share information, watch videos, and build online communities. The success of these platforms depends on effective information architecture, navigation design, and interaction patterns. These design principles help users find relevant information quickly and encourage continued engagement. However, poorly designed interfaces may cause confusion, distraction, and cognitive overload.

1. Information architecture

Information architecture (IA) refers to the systematic organization, classification, and presentation of information within an application.

a) Content categorization

Social media platforms organize content into categories such as posts, reels, videos, stories, messages, groups, and notifications. Proper categorization helps users understand where information is located.

b) Hierarchical organization

Important features are placed prominently, while secondary features are organized within menus or submenus. For example, the home feed is usually more prominent than settings.

c) Search and filtering

Search bars, hashtags, keywords, and filters allow users to locate specific people, posts, videos, and topics.

d) Personalized content organization

Algorithms recommend posts based on users' interests, viewing history, and interactions. Personalization can improve content relevance, but it can also create repetitive content and filter bubbles.

e) Consistency

Consistent labels, icons, and layouts across different screens help users learn the interface more easily.

2. Navigation paradigms

Navigation paradigms define how users move between sections and access information.

a) Tab-based navigation

A bottom or top navigation bar provides access to important sections, such as Home, Search, Reels, Notifications, and Profile.

b) Feed-based navigation

Users scroll vertically through a continuous stream of content. Infinite scrolling reduces interruptions but may encourage excessive use.

c) Search-based navigation

Users enter keywords, names, or hashtags to locate specific content directly.

d) Hierarchical navigation

Users move through levels such as Home → Groups → Group Details → Posts. Breadcrumbs or back buttons can help users understand their location.

e) Notification-based navigation

Notifications provide direct links to comments, likes, messages, and other activities. They help users return to relevant content but can also interrupt ongoing tasks.

3. Interaction design patterns

Interaction design patterns are reusable solutions to common interface problems.

a) Like, comment, and share

These familiar controls make interaction simple and encourage users to participate in discussions.

b) Follow and subscribe

Users can receive updates from people, creators, or channels they find interesting.

c) Infinite scrolling

New content loads automatically as users move down the feed. This reduces navigation effort but can make it difficult to stop browsing.

d) Pull-to-refresh

Users can refresh a feed through a simple gesture to view updated content.

e) Notifications and feedback

Badges, animations, sound alerts, and confirmation messages communicate the result of user actions.

f) Personalization

Recommendation systems present content based on user interests and previous behaviour, increasing the likelihood of relevant engagement.

4. Impact on user engagement

Effective information architecture and interaction patterns improve user engagement in several ways:

1. Ease of use: Clear categories and navigation reduce the effort required to find content.
2. Increased participation: Likes, comments, shares, and messaging make interaction easy.
3. Personal relevance: Recommendations help users discover content related to their interests.
4. Social connection: Notifications and messaging support communication with friends and communities.
5. User retention: Consistent design and useful features encourage users to return to the platform.

However, engagement should not be measured only by the amount of time users spend on a platform. Meaningful interaction, satisfaction, accessibility, and user control are also important.

5. Cognitive overload

Cognitive overload occurs when the amount of information or number of decisions exceeds the user's available mental processing capacity.

Social media can cause cognitive overload through:

- Excessive notifications and interruptions.
- Large quantities of rapidly changing content.
- Too many choices and competing visual elements.
- Repeated recommendations and advertisements.
- Frequent switching between messages, feeds, and videos.

- Infinite scrolling that makes it difficult to estimate how much content remains.

These factors can reduce concentration, increase mental fatigue, and make important information difficult to remember.

6. Design solutions to reduce cognitive overload

Designers can reduce cognitive overload by applying the following principles:

1. Visual hierarchy: Use headings, spacing, size, and contrast to distinguish important information.
2. Progressive disclosure: Display essential information first and reveal advanced options only when required.
3. Notification control: Allow users to disable, group, or schedule notifications.
4. Clear navigation: Use familiar labels, consistent icons, and a manageable number of menu options.
5. User control: Provide controls for recommendation preferences, feed settings, and autoplay.
6. Accessibility: Support readable text, sufficient contrast, screen readers, and keyboard navigation.
7. Meaningful feedback: Communicate successful actions without unnecessary animations or interruptions.

15. Show how affordances, signifiers, mapping, feedback, and constraints are systematically implemented to design an intuitive multi-functional kitchen appliance control panel.

Introduction

A multi-functional kitchen appliance, such as a microwave oven, electric cooker, or combination oven, may provide several functions, including heating, grilling, defrosting, timer settings, and temperature control. If the control panel is poorly designed, users may select incorrect functions, enter inappropriate settings, or operate the appliance unsafely.

The principles of affordances, signifiers, mapping, feedback, and constraints help designers develop a control panel that is easy to understand, efficient to use, and safe.

1. Affordances

Affordances are the possible actions that an object or interface allows a user to perform.

In a kitchen appliance control panel, different controls should support the actions expected of them.

Implementation:

- A physical knob affords turning.
- A raised button affords pressing.
- A touchscreen slider affords dragging.
- A door handle affords pulling.
- A Start button affords initiating a cooking operation.

Each control should have a shape and behaviour appropriate to its function. For example, a rotary knob is suitable for adjusting temperature or cooking time because it allows continuous adjustment.

Example: A large rotary knob can be used to set cooking time, while a clearly pressable button starts the operation.

Good affordances reduce the need for lengthy instructions and make the appliance easier to learn.

2. Signifiers

Signifiers are visible or perceivable indicators that communicate where and how an action should be performed. An affordance describes a possible action, whereas a signifier helps the user discover that action.

Implementation:

- Label buttons as Start, Stop, Timer, and Temperature.
- Use recognizable symbols for microwave, grill, defrost, and fan functions.
- Provide arrows around rotary knobs to indicate the direction of adjustment.
- Use contrasting colours to distinguish important controls.
- Display units such as °C, minutes, and seconds beside the corresponding values.

Example: A snowflake symbol accompanied by the label “Defrost” helps users identify the defrosting function.

Labels should be clear and consistent. Designers should not rely exclusively on colour or unfamiliar icons because users may have different levels of experience or visual ability.

3. Mapping

Mapping refers to the relationship between a control and the result it produces. Good mapping makes the relationship between the user's action and the appliance's response easy to understand.

Implementation:

- Arrange temperature controls near the temperature display.
- Place timer controls close to the cooking-time display.

- Use a clockwise knob movement to increase time and an anticlockwise movement to decrease it, with clear directional indicators.
- Arrange temperature values in increasing order.
- Place Start and Stop controls in predictable locations.

Example: If a user turns a temperature knob clockwise, the displayed temperature should increase. If the user turns it anticlockwise, the temperature should decrease.

Poor mapping can cause errors. For example, placing the temperature and timer controls far apart from their displays may confuse users about which setting they are changing.

4. Feedback

Feedback communicates the result of a user's action. It confirms whether the system has received and processed an instruction.

Implementation:

1. Display the selected cooking mode and temperature.
2. Show a countdown timer during operation.
3. Provide a sound or visual signal when cooking finishes.
4. Illuminate the selected function.
5. Display an error message if the door is open or a setting is invalid.
6. Show a clear confirmation when a setting is saved.

Example: When the user presses Start, the display should show the remaining cooking time and indicate that cooking has begun.

Feedback should be immediate, clear, and appropriate. Important warnings must be distinguishable from ordinary status messages.

5. Constraints

Constraints limit the actions a user can perform to prevent mistakes and unsafe operation.

Implementation:

- Prevent operation when the door is open.
- Restrict temperature settings to supported values.
- Prevent negative cooking times.
- Disable incompatible functions when another mode is selected.
- Require confirmation before cancelling a long cooking cycle when appropriate.
- Provide a child lock to prevent accidental operation.

Example: If the user tries to start a microwave while the door is open, the appliance should refuse to start and display “Please close the door.”

Constraints can be physical, logical, semantic, or cultural. Physical constraints prevent actions through the structure of the device, while logical constraints prevent invalid combinations of settings.

6. Systematic integration of the principles

These principles should work together rather than independently.

Consider the process of heating food:

1. The user identifies the Microwave button through its signifier.
2. The button affords pressing, providing an affordance.
3. The user selects a power level and turns the time knob; the controls have clear mapping to the displayed settings.
4. The screen immediately displays the selected power and cooking time, providing feedback.
5. If the door is open or the selected value is invalid, constraints prevent the operation.
6. After the user presses Start, the countdown display and completion alarm communicate the progress and result.

7. Additional design considerations

Consistency: Similar controls should behave similarly across different operating modes.

Error prevention: Provide reasonable defaults and prevent incompatible settings.

Accessibility: Use large buttons, readable text, high contrast, and tactile controls where appropriate.

Safety: Separate dangerous or irreversible operations from ordinary controls, and provide clear warnings.

Simplicity: Group related functions together and avoid unnecessary controls.

Usability testing: Ask representative users to perform common tasks, such as defrosting food or setting a timer. Observe errors, task completion time, and user satisfaction, then improve the design.

16. Illustrate the complete interaction design and spatial user interface (3D UI) architecture for an Augmented Reality (AR) surgical training and anatomical visualization system.

Introduction

Augmented Reality (AR) is a technology that combines computer-generated information with the real-world environment. In medical education, AR can help students and surgeons understand human anatomy, practise surgical procedures, and visualize internal organs in three dimensions.

An AR surgical training and anatomical visualization system overlays virtual anatomical models onto a real patient model, mannequin, or physical training environment. Users can interact with these models through hand gestures, voice commands, touch controls, or tracked instruments.

The system must provide accurate spatial alignment, realistic visualization, responsive interaction, clear feedback, and safe training conditions.

1. System architecture

The architecture of an AR surgical training system consists of several connected components.

User and physical environment

Surgeon, student, mannequin, physical instruments

Sensing and tracking layer

Cameras, depth sensors, hand tracking, instrument tracking

AR processing and registration layer

Spatial mapping, coordinate transformation, pose estimation

Anatomical and simulation engine

3D models, tissue properties, surgical simulation, assessment

Rendering and interaction layer

3D visualization, menus, labels, gestures, voice, feedback

AR display and user feedback

Head-mounted display, spatial overlays, visual and audio feedback

The system operates as a continuous loop. The sensors detect user movement, the processing layer updates the position of virtual objects, and the rendering layer displays the updated scene.

2. Anatomical data and 3D modelling

The system uses medical imaging data, such as CT scans and MRI scans, to construct three-dimensional anatomical models.

Important features include:

- Three-dimensional representations of organs, bones, muscles, nerves, and blood vessels.
- Separate anatomical layers that can be shown or hidden.
- Colour coding to distinguish different structures.
- Labels and annotations identifying anatomical parts.
- Models that support rotation, zooming, slicing, and cross-sectional views.

Example: A medical student can view a 3D heart model, rotate it, and examine its chambers and blood vessels from different angles.

The anatomical models should be reviewed by qualified medical professionals to ensure educational accuracy.

3. Spatial user interface design

A spatial user interface places virtual controls and information within the user's three-dimensional environment rather than limiting the interface to a flat screen.

a) Spatial anchoring

Virtual anatomical structures should remain aligned with the physical mannequin or training model as the user moves.

b) Depth and positioning

Important structures should appear at appropriate distances and depths to support accurate perception.

c) Direct manipulation

Users should be able to select, rotate, enlarge, or isolate anatomical structures using natural gestures or tracked tools.

d) Contextual menus

Controls should appear near the relevant anatomical structure when useful, without blocking the user's view.

e) Layer management

Users should be able to reveal or hide structures such as skin, muscles, bones, and organs.

f) Visual hierarchy

Important structures should be highlighted without overwhelming the user with labels, animations, or unrelated information.

4. Interaction techniques

The system should support multiple interaction methods to suit different training activities.

Hand gestures: Pinching can select a structure, while dragging can rotate or reposition a virtual model.

Voice commands: Commands such as “Show the heart” or “Hide muscles” allow hands-free interaction.

Gaze interaction: Looking at a structure can highlight it or prepare it for selection.

Instrument tracking: Specially tracked surgical instruments can interact with virtual tissues in the training environment.

Touch interaction: A connected tablet or control panel can be used to select procedures, change settings, or review instructions.

For safety and accuracy, the system should provide clear confirmation when an important action is performed and prevent accidental changes during a training task.

5. Surgical simulation and training

A surgical training module allows students to practise selected procedures in a simulated environment.

The system may include:

1. Guided training: Step-by-step instructions demonstrate the correct sequence of actions.
2. Anatomical exploration: Students inspect the location and relationship of anatomical structures.
3. Instrument simulation: The system tracks virtual or physical instruments.
4. Procedure guidance: Visual indicators identify the target area and important structures.
5. Error detection: The system identifies deviations from the predefined training procedure.
6. Performance evaluation: The system records task completion time, movement accuracy, and procedural errors.

For example, during a simulated procedure, the system can highlight a target anatomical region and warn the trainee when a tracked instrument approaches a designated structure that should be avoided.

Such feedback must be based on validated training rules rather than being presented as an authoritative clinical decision.

6. Feedback and error prevention

Feedback is important because users need to understand how their actions affect the virtual environment.

The system can provide:

- Visual highlights when an anatomical structure is selected.
- Labels confirming the selected organ.
- Audio notifications for important training events.
- Progress indicators showing the current training stage.
- Warnings when the user deviates from a simulated procedure.
- A reset option to restore the initial training state.

Error-prevention mechanisms should include confirmation for resetting a session, protection against accidental deletion of progress, and clear instructions when tracking is lost.

The system should never silently display inaccurate spatial alignment. If tracking becomes unreliable, it should clearly notify the user and suspend any affected precision-dependent training task.

7. Usability, accessibility, and safety

The system should be designed to reduce fatigue and mental workload.

Usability: Keep frequently used controls accessible and provide a consistent interaction pattern.

Accessibility: Support readable labels, sufficient contrast, adjustable text sizes, and alternatives to gesture-based interaction.

Comfort: Reduce unnecessary visual movement and provide options to adjust the viewing arrangement.

Safety: Distinguish simulated information from real patient data and prevent users from confusing educational overlays with validated clinical guidance.

Privacy: Protect medical images and any personal information used during training.

Performance: Maintain low interaction latency and reliable tracking so that virtual structures remain aligned with the physical environment.

8. Evaluation and testing

The system should be tested with medical students, instructors, and relevant clinical experts.

Evaluation criteria include:

- Accuracy of anatomical visualization and spatial registration.
- Time taken to complete training tasks.
- Number and severity of interaction errors.
- Ease of learning and navigation.
- User satisfaction and perceived workload.
- Stability of tracking and responsiveness.
- Effectiveness of training compared with appropriate existing methods.

Testing should be performed in a controlled training environment, and the system should be improved based on observed problems and expert feedback.